

CASE STUDY: SYSTEMIC QUALITY TRANSFORMATION

Targeting \$1.6M in Non-Recoverable Site Cost Reductions

Executive Summary A strategic intervention to transition a façade production system from reactive defect management to a preventive quality governance model. By identifying upstream root causes, annual non-recoverable costs were reduced from **\$2M to \$400K**, delivering a total savings of **\$1.6M**.

1. CONTEXT & PROBLEM STATEMENT

The organization faced recurring site non-conformances (NCRs), including surface scratches, finishing defects, and installation errors. These issues led to excessive rework, replacements, and high labor costs.

- **Financial Impact:** ~\$2M annual non-recoverable loss.
- **Root Cause Identification:** Data analysis across 40+ sites revealed that **60% of site issues** originated during the production and handling stages, rather than at the point of installation.
- **Systemic Gaps:** Absence of standardized end-of-line inspection.
 - Lack of clear quality ownership.
 - Disconnected feedback loop between site and production teams.

2. MANAGEMENT APPROACH (SYSTEMS THINKING)

Applying a framework-driven approach to treat the failure as a system-wide deficiency rather than isolated technical errors.

- **RCA (Root Cause Analysis):** Utilized **5 Whys** and **Fishbone (Ishikawa)** diagrams to separate symptoms from systemic drivers.
- **Gemba Strategy:** Conducted on-site shop floor observations to identify gaps in real-time handling and finishing practices.
- **Stakeholder Alignment:** Bridged the communication gap between Production and Site Installation teams.

3. STRATEGIC SOLUTION & IMPLEMENTATION

I led a cross-functional team to implement a Preventive Quality Control System:

1. **SOP Standardization:** Developed and enforced rigorous end-of-line cleaning and inspection Standard Operating Procedures.
2. **Governance & Ownership:** Established a "Final QC Sign-off" protocol, assigning clear accountability for product integrity prior to shipment.
3. **Capability Building:** Conducted training for production teams on defect prevention and handling standards.

4. **Feedback Loop:** Automated a site-to-production reporting mechanism to ensure recurring issues were closed via **8D methodology**.

4. QUANTIFIED IMPACT (ROI)

Key Performance Indicator	Baseline (Pre-Intervention)	Post-Intervention Result	% Improvement
Annual Site Costs	\$2,000,000	\$400,000	80% Reduction
Total Savings	\$0	\$1,600,000	N/A
Site Quality Rating	Reactive / High Risk	Stable / Preventive	Major Impact

5. LEADERSHIP TAKEAWAYS

- **Prevention vs. Reaction:** Preventive quality systems eliminate the "hidden factory" of rework costs.
- **Upstream Accountability:** Sustainable performance is achieved when quality is owned at the source.
- **Standardization:** Consistent SOPs are the foundation of operational excellence and scalable cost reduction.

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